

ARBIN CHAUDHARY

Geomatics Engineer

+977-9866202247 | arbinchaudhary.0006@gmail.com | linkedin.com/in/arbin-chaudhary-a8a16931b
arbinchaudhary.com.np | Ghorahi-9, Dang, Nepal

PROFESSIONAL SUMMARY

Recent Geomatics Engineering graduate (Kathmandu University) with comprehensive hands-on training in surveying technologies, GIS spatial analysis, and environmental modeling. Field experience includes hydropower, bridge, and route surveys using total station; UAV-derived DEM processing; and multi-criteria site selection using AHP in ArcGIS. Thesis research combined machine learning with HEC-RAS for flood inundation mapping. Committed to applying geomatics expertise in dynamic, mission-driven engineering environments across Nepal and internationally.

EDUCATION

Bachelor of Engineering in Geomatics Engineering

Kathmandu University, Nepal

2021 – 2026

Graduated 2025/26

Relevant Coursework: Surveying & Mapping Technologies, Geographic Information Systems (GIS), Geospatial Database Management, Remote Sensing & Spatial Analysis, GNSS Surveying, Cadastral Survey & Land Administration, Hydraulic Modeling & Environmental Assessment

Higher Secondary (+2 Science)

KIST College & Secondary School, Kamalpokhari, Kathmandu

Completed 2021

Secondary Level Education (Grades 1–10)

Shanti Sudha Secondary School, Ghorahi-9, Dang

Completed 2015

TECHNICAL SKILLS

Surveying & Field Ops	Theodolite (topographic mapping, traversing, triangulation), Auto Level, Total Station, Plane Table Surveying, GNSS/GPS, UAV/Drone Mapping, Differential Leveling
GIS & Spatial Analysis	Multi-Criteria Decision Making (MCDM), AHP, Thematic Mapping, Digital Elevation Model (DEM), Land Use/Land Cover Classification, Network Analysis
Hydrological Modeling	Flood Inundation Mapping, Watershed Analysis, Terrain Modeling, UAV-Derived DEM Processing
Software	ArcGIS Desktop, QGIS, HEC-RAS, Google Earth Engine, AutoCAD
Programming	Python, Java, SQL (Spatial Databases), HTML/CSS/JavaScript, Leaflet.js, Turf.js

ACADEMIC PROJECTS & FIELD SURVEYS

Machine Learning & HEC-RAS Integrated Flood Inundation Mapping — Bagmati River, Khokana

2025/26

8th Semester Thesis — Geospatial Engineering, Hydraulic Modeling, Machine Learning

- Developed integrated ML and HEC-RAS models to simulate and map flood inundation extents along the Bagmati River corridor near Khokana, Nepal.
- Combined remote sensing data, DEM analysis, and hydraulic simulation for high-accuracy flood hazard delineation.

Comprehensive Field Survey — Cadastral, GNSS, UAV & Precise Levelling

2025/26

8th Semester Field Survey

- Executed traverse survey for digital cadastral mapping, plane table cadastral survey, GNSS positioning, UAV photogrammetric mapping, and precise levelling operations.

Lumbini Province Tourist Explorer Web Application

2024/25

7th Semester Project — HTML, CSS, JavaScript, Leaflet.js, Turf.js

- Built an interactive web mapping application for tourism in Lumbini Province integrating multiple base maps, spatial queries, and geospatial visualizations.

- Designed and deployed a professional portfolio at arbinchaudhary.com.np to showcase geomatics projects and technical skills.

Total Station Survey — Hydropower, Bridge & Route Survey

2024/25

6th Semester Field Survey — Infrastructure Engineering

- Hydropower:** Established instrument stations, measured angles/distances for penstock alignment, determined elevation changes for water head calculations, and generated cross-sections for capacity assessment.
- Bridge:** Measured structural dimensions, established survey control points referenced to national datum, and created bridge deck profiles and geometric documentation.
- Route Survey:** Surveyed proposed corridors, determined optimal alignment from topographic and engineering constraints, and collected longitudinal and cross-sectional profiles.

Thematic Mapping & Spatial Visualization

2024/25

6th Semester Mini Project — ArcGIS, QGIS

- Produced thematic maps including land use/land cover, slope and aspect, drainage and watershed boundaries, soil/geological formations, and population and infrastructure distribution.

Shortest Route Optimization — QGIS Network Analysis

2023/24

5th Semester Mini Project

- Applied network analysis algorithms in QGIS to compute optimal routes for transportation planning scenarios.

Comparative Cadastral Survey Study — DGPS/GNSS vs. Total Station

2023/24

5th Semester

- Conducted comparative analysis of DGPS/GNSS and Total Station methodologies for cadastral surveying, evaluating accuracy, efficiency, and applicability.

GIS-Based MCDM for Landfill Site Selection — Dang District

2022/23

4th Semester Mini Project — ArcGIS, Spatial Analysis

- Used GIS-based multi-criteria decision making to identify suitable landfill locations, integrating environmental, socioeconomic, and infrastructural spatial criteria.

Theodolite Field Surveys — Leveling, Traversing, Triangulation & Topographic Mapping

2022/23

3rd & 4th Semester Field Surveys

- Performed auto-level differential leveling, closed and open traverses, triangulation networks, and full topographic mapping using theodolite equipment.

LANGUAGES

Nepali — Native

Tharu — Mother Tongue

English — Fluent

Hindi — Conversational

CORE COMPETENCIES

- Strong analytical and problem-solving skills
 - Effective communication and team collaboration
 - Precision and attention to detail in data collection
- Adaptability to emerging technologies and tools
 - Time management across concurrent engineering projects
 - Critical thinking for complex engineering decisions

CAREER OBJECTIVE

Seeking to leverage geomatics engineering expertise in challenging roles involving infrastructure planning, environmental site assessment using MCDM, hydrological and flood risk modeling, UAV and remote sensing applications, and the integration of machine learning with geospatial methodologies — contributing to sustainable development and data-driven engineering decisions at national and international levels.